Implementation of a Multi-Modal Real-Time Video Communication System for Speech and Hearing-Impaired Individuals

CAPSTONE PROJECT REPORT

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DECLARATION

We affirm that the project work titled "**Design and Implementation of a Multi-Modal Real-Time Video Communication System for Speech and Hearing-Impaired Individuals**" being submitted in partial fulfilment for the award of the degree of **Bachelor of Technology in Computer Science and Engineering** is the original work carried out by us. It has not formed part of any other project work submitted for the award of any degree or diploma, either in this or any other University.

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BONAFIDE CERTIFICATE

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PROJECT SUMMARY

Project Title	Design and Implementation of a Multi-Modal Real-Time Video Communication System for Speech and Hearing-Impaired Individuals	
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Program Concentration Area	Artificial Intelligence, Computer Vision, and Natural Language Processing	
Technical Requirements	Machine Learning, Deep Learning, Web Development	
Engineering standards and realistic constraints in these areas		
Area	**Codes & Standards / Realistic Constraints**	**Tick ✓**
Economic	The proposed system is designed to be cost-effective by utilizing open-source frameworks and scalable cloud-based deployment. It eliminates the need for human interpreters, reducing long-term communication costs in institutions and workplaces. Additionally, the use of standard hardware such as webcams and microphones ensures affordability and easy adoption.	✓
Social	The system promotes inclusivity by enabling seamless communication between deaf and hearing individuals in real-time. It enhances accessibility in education, workplaces, and public services, reducing communication barriers.	✓

ABSTRACT

In the evolving landscape of assistive technologies, the integration of artificial intelligence into communication systems has significantly improved accessibility for individuals with hearing and speech impairments. This report presents the design and implementation of a Multi-Modal Real-Time Video Communication System that enables seamless translation between speech, text, and sign language within a unified platform. The proposed system leverages computer vision techniques for real-time sign language recognition and natural language processing for speech-to-text and text-to-speech conversion. By incorporating deep learning models such as Spatio-Temporal Graph Convolutional Networks (ST-GCN), along with Automatic Speech Recognition and avatar-based sign rendering, the framework ensures accurate and synchronized bi-directional communication. The system operates within a WebRTC-based video conferencing environment, enabling low-latency interaction and real-time captioning. Performance evaluation demonstrates high recognition accuracy and reliable real-time responsiveness when compared to existing single-function assistive tools. By integrating multiple AI modules into a cohesive architecture, the proposed solution enhances digital inclusivity and simplifies communication between deaf and hearing individuals across educational, professional, and social environments.

Keywords – SIGN LANGUAGE RECOGNITION, REAL-TIME TRANSLATION, COMPUTER VISION, NATURAL LANGUAGE PROCESSING, ST-GCN, AUTOMATIC SPEECH RECOGNITION, WEBRTC, ACCESSIBILITY TECHNOLOGY, MULTI-MODAL COMMUNICATION

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LIST OF ACADEMIC REFERENCE COURSES

S. NO.	COURSE CODE	COURSE NAME
1	212CSE1402	INTRODUCTION TO PYTHON PROGRAMMING
2	212CSE2303	PREDICTIVE ANALYTICS
3	213CSE2304	MACHINE LEARNING
4	231CSE3301	DEEP LEARNING
5	213CSE2303	SOFTWARE ENGINEERING

CHAPTER – I
INTRODUCTION

1.1 Overview:

The rapid advancement of artificial intelligence, computer vision, and real-time communication technologies has accelerated the development of intelligent systems designed to enhance accessibility and inclusivity. Among these advancements, assistive communication platforms have emerged as a critical solution for bridging the communication gap between deaf, hard-of-hearing, and hearing individuals. With increasing global emphasis on digital inclusion and equal access to communication tools, the demand for real-time, AI-powered translation systems continues to grow across educational institutions, workplaces, healthcare facilities, and social environments.

In response to these challenges, this project proposes a novel **Multi-Modal Real-Time Video Communication System** that integrates sign language recognition, speech-to-text conversion, text-to-speech synthesis, and text-to-sign translation within a unified video conferencing platform. Unlike traditional assistive tools that operate independently—focusing solely on captioning or sign recognition—the proposed framework enables seamless, bi-directional communication between users relying on speech, text, or sign language.

At the core of the system lies an intelligent recognition engine powered by advanced deep learning architectures such as Spatio-Temporal Graph Convolutional Networks (ST-GCN), Convolutional Neural Networks (CNN), and recurrent models. These models process real-time video input to accurately recognize sign language gestures and translate them into textual and spoken output. Simultaneously, speech recognition technologies such as Whisper-based Automatic Speech Recognition (ASR) convert spoken language into text, which can then be rendered as natural speech or animated sign language through a 3D avatar interface.

The system leverages WebRTC-based real-time streaming, ensuring synchronized transmission of audio, video, captions, and sign animations with minimal latency. A scalable backend architecture built using Node.js, Python microservices, and containerized deployment ensures robustness and adaptability across various environments.

By integrating multiple AI modules into a single, cohesive platform, the proposed framework addresses the limitations of fragmented assistive technologies. The solution is designed to enhance accessibility in diverse settings, including education, professional communication, telemedicine, and remote collaboration. Ultimately, this project aims to promote digital inclusivity by enabling natural, real-time, two-way communication between deaf and hearing communities, redefining accessibility standards in modern communication systems.

1.2 Use Cases & Applications:

The proposed multi-modal communication system can be applied across various domains where inclusive and real-time interaction is essential. In educational institutions, it enables seamless communication between deaf students and instructors during live classes and online sessions. In corporate environments, it supports accessible meetings, interviews, and team collaborations by providing instant speech-to-sign and sign-to-speech translation. In healthcare settings, the system facilitates effective communication between medical professionals and patients with hearing impairments, reducing dependency on interpreters and improving service efficiency. Additionally, it enhances accessibility in public services, customer support centres, and everyday personal video calls. Overall, the framework promotes digital inclusivity by enabling real-time, two-way communication between deaf and hearing individuals across diverse real-world scenarios.

1.3 Challenges:

Despite the strong potential of the proposed multi-modal communication system, several challenges must be addressed to ensure accuracy, reliability, and real-world usability. One of the primary challenges is achieving high-precision sign language recognition in diverse environments. Variations in lighting conditions, camera angles, background noise, and user-specific signing styles can significantly affect gesture detection and classification accuracy. Developing robust deep learning models capable of handling continuous and dynamic gestures remains a critical requirement. Another major challenge lies in maintaining low-latency, real-time communication. Since the system integrates multiple AI modules—sign recognition, speech recognition, text-to-speech, and avatar rendering—ensuring synchronized processing within a live WebRTC video stream is technically demanding. Any delay in processing can disrupt the natural flow of conversation. Additionally, dataset limitations pose a significant hurdle. Sign languages vary regionally, and large, well-annotated datasets for languages such

as Indian Sign Language (ISL) are limited. This restricts model generalization and sentence-level translation performance. Finally, scalability and computational efficiency are important considerations. Deploying deep learning models in real-time environments requires optimization techniques to ensure smooth performance across different devices and network conditions. Addressing these challenges is essential to delivering a reliable, inclusive, and widely adoptable communication platform.

CHAPTER-II

LITERATURE REVIEW

Developing intelligent systems capable of enabling seamless communication between deaf and hearing individuals has attracted significant research interest in recent years. Sign language recognition and real-time speech translation remain complex challenges due to variations in gestures, environmental conditions, and computational constraints. The proposed system addresses these challenges by integrating computer vision, deep learning, natural language processing, and WebRTC-based real-time communication into a unified platform. The following literature highlights key technologies relevant to the proposed framework.

2.1 Deep Learning for Sign Language Recognition

In the study "Real-time American Sign Language interpretation using deep learning and key point tracking" (El-Sayed et al., 2022), a MediaPipe-based key point extraction method combined with deep learning models was used for ASL recognition. The system achieved strong performance but required large datasets and faced challenges in continuous gesture recognition. Similar approaches are incorporated in the proposed system, where key point detection is used as input to advanced models such as ST-GCN for improved spatial-temporal understanding.

2.2 CNN and RNN-Based Translation Systems

Agre et al. (2023) proposed a bi-directional translation system using CNN and RNN architectures to convert sign language to text and speech, and vice versa. While the study demonstrated feasibility in two-way communication, it was limited to specific datasets and language scopes. The proposed framework extends this approach by integrating multiple deep learning models (CNN, LSTM, GRU, ST-GCN) and embedding them within a real-time video conferencing environment.

2.3 WebRTC-Based Accessible Communication Platforms

Kumar et al. (2023) developed a WebRTC-based video chat system integrated with sign language detection. Their system demonstrated that accessibility features can be directly embedded within video conferencing platforms. However, performance was sensitive to lighting conditions and bandwidth limitations. The proposed system improves upon this by optimizing latency and synchronizing speech, captions, and 3D avatar rendering for smoother interaction.

2.4 Spatio-Temporal Modelling for Gesture Recognition

Huang and Chouvatut (2024) utilized ResNet combined with LSTM networks for video-based sign language recognition. Their approach highlighted the importance of capturing both spatial and temporal dependencies in gesture sequences. Similarly, the proposed system employs Spatio-Temporal Graph Convolutional Networks (ST-GCN), which more effectively model skeletal key points over time, resulting in improved real-time recognition accuracy.

2.5 Text-to-Sign and Avatar-Based Systems

Mehta et al. (2019) introduced automated 3D sign language avatars for caption generation and speech conversion. While the system enhanced accessibility, it raised concerns regarding avatar naturalness and realism. The proposed framework integrates text-to-sign avatar rendering within a live communication pipeline, aiming to improve synchronization and usability in practical environments.

2.6 Comparison with Existing Approaches

Unlike prior research that focuses on single-function tools such as sign-to-text recognition, captioning systems, or avatar generation independently, the proposed system integrates sign recognition, speech-to-text, text-to-speech, and text-to-sign translation within a unified real-time video communication framework. While many earlier systems emphasize navigation of gestures or isolated translation modules, the proposed framework prioritizes end-to-end, bi-directional communication with low latency and scalable deployment.

By combining computer vision, deep learning architectures, NLP-based speech processing, and WebRTC streaming, the proposed system aims to overcome fragmentation in existing assistive technologies and deliver a comprehensive, real-time, and inclusive communication platform.

2.7 Automatic Speech Recognition (ASR) in Real-Time Systems

Recent advancements in Automatic Speech Recognition (ASR) have significantly improved real-time speech-to-text conversion accuracy. Models such as transformer-based ASR systems (e.g., Whisper) have demonstrated strong performance across diverse accents and noisy environments. However, real-time deployment in live communication systems introduces latency and synchronization challenges. Many studies focus only on speech transcription without integrating sign translation. The proposed system incorporates ASR within a real-time video communication pipeline, enabling speech-to-text conversion that is directly synchronized with sign avatar rendering for seamless interaction.

2.8 Graph-Based Deep Learning for Skeleton Modelling

Graph-based neural networks have emerged as powerful tools for modelling structured data such as human skeletal key points. Research on Spatio-Temporal Graph Convolutional Networks (ST-GCN) has shown superior performance in action recognition and gesture classification tasks by capturing both spatial joint relationships and temporal motion patterns. Unlike traditional CNN or LSTM-based approaches that process frames sequentially, ST-GCN models joint dependencies more effectively. The proposed system adopts ST-GCN to enhance continuous sign recognition accuracy and improve robustness in dynamic real-world scenarios.

2.9 Cloud-Based and Microservice Architectures for AI Deployment

Several recent studies highlight the importance of scalable backend architectures for deploying AI models in real-time applications. Cloud-based processing and microservice architectures enable modular design, efficient load handling, and easier model updates. However, reliance on cloud services may introduce latency and bandwidth constraints. The proposed system balances this trade-off by integrating Python-based ML microservices with containerized

deployment, ensuring scalable, real-time inference while maintaining low latency within a WebRTC-based communication framework.

CHAPTER 3
PROBLEM DEFINITION & BACKGROUND

3.1 Problem Definition:

The proposed system aims to address critical challenges in real-time communication between deaf, hard-of-hearing, and hearing individuals by integrating multiple AI-driven components within a unified platform. The primary challenge lies in enabling seamless, bi-directional translation between sign language, speech, and text during live video interactions. The system leverages computer vision techniques for real-time sign language recognition, where skeletal key points extracted from video streams are processed using deep learning models. Simultaneously, Automatic Speech Recognition (ASR) modules convert spoken language into text, which can then be synthesized into speech or rendered as sign animations through a 3D avatar interface. A WebRTC-based communication layer orchestrates synchronized transmission of video, audio, captions, and sign animations, ensuring minimal latency and smooth interaction. The backend architecture integrates machine learning microservices, database management, and scalable deployment mechanisms to support real-time inference and continuous communication. By combining gesture recognition, speech processing, and avatar rendering into a single framework, the system enhances accessibility and autonomy, eliminating the need for external interpreters or fragmented assistive tools.

3.2 Problem Formulation:

The proposed system, denoted as Framework C, aims to facilitate real-time, bi-directional communication between users operating in different communication modes within an interactive environment E.

$$C = \{c_1, c_2, c_3, c_4\} \in E$$

Where:

- c_1 represents the Sign Language Recognition module, responsible for extracting skeletal key points from live video input and classifying gestures into textual representations using deep learning architectures such as ST-GCN.

- c_2 represents the Automatic Speech Recognition (ASR) module, which converts live audio streams into text transcripts.
- c_3 represents the Text-to-Speech (TTS) module, generating natural voice output from recognized or translated text.
- c_4 represents the Text-to-Sign Translation module, which converts textual input into animated sign language representations using pre-recorded sign sequences or 3D avatars.

The environment E denotes real-time video communication settings, including educational platforms, workplaces, healthcare environments, and personal communication spaces.

The objective of Framework C is to optimize accuracy, latency, and synchronization across all sub-modules, enabling smooth, natural, and inclusive communication. By ensuring efficient interaction between visual, auditory, and textual modalities, the system seeks to reduce communication barriers and promote digital accessibility across diverse real-world environments.

CHAPTER 4
PROPOSED SYSTEM

The proposed framework aims to develop a Multi-Modal Real-Time Communication System that enables seamless interaction between deaf, hard-of-hearing, and hearing individuals. The system integrates computer vision, deep learning, natural language processing, and WebRTC-based video communication into a unified platform. It performs real-time sign language recognition, speech-to-text conversion, text-to-speech synthesis, and text-to-sign avatar rendering within a single communication environment. The results are displayed live within the video conferencing interface, allowing users to communicate naturally without relying on external interpreters.

4.1 Video and Key point Capture Integration

The system utilizes a real-time video stream to capture sign language gestures. A key point extraction framework such as MediaPipe or OpenPose detects skeletal landmarks of the hands, face, and upper body. These extracted key points form structured input data representing spatial relationships between joints. The processed key point sequences are forwarded to the deep learning model for gesture recognition. This approach ensures reduced computational load compared to raw image processing while maintaining high recognition accuracy.

4.2 Integration of Deep Learning and NLP Modules

The proposed framework incorporates two essential AI modules: a **Sign Recognition Module** and a **Speech Processing Module**.

The sign recognition module utilizes deep learning architectures such as CNN, LSTM, GRU, and Spatio-Temporal Graph Convolutional Networks (ST-GCN). These models are trained on sign language datasets to classify gestures into words or sentence-level outputs.

The speech processing module integrates Automatic Speech Recognition (ASR) systems such as Whisper to convert spoken language into text. The generated text is then processed by Text-to-Speech (TTS) engines or converted into sign animations using a 3D avatar. The integration of these modules ensures bi-directional translation within the same platform.

4.3 Sign Recognition Algorithm Deployment

The Spatio-Temporal Graph Convolutional Network (ST-GCN) plays a crucial role in modelling gesture sequences. Unlike traditional CNN-based approaches that analyse individual frames independently, ST-GCN captures both spatial relationships between skeletal joints and temporal dependencies across multiple frames.

The model processes key point graphs over time to classify dynamic gestures accurately. By leveraging spatial-temporal feature extraction, the system achieves improved performance in continuous sign recognition. The trained model is deployed as a backend microservice, enabling real-time inference during live video communication.

4.4 Real-Time Video Communication using WebRTC

The framework integrates WebRTC technology for peer-to-peer video and audio streaming. WebRTC ensures low-latency communication and synchronized transmission of video, audio, captions, and avatar animations. A Node.js-based signalling server manages session establishment, while backend APIs handle model inference and data exchange. This integration enables smooth and uninterrupted two-way conversations.

4.5 Text-to-Sign Avatar Rendering

To support hearing users communicating with sign language users, the system includes a text-to-sign translation module. Recognized speech transcripts are converted into animated sign sequences using pre-recorded sign clips or a 3D avatar framework. Facial expressions and hand movements are synchronized with text output to improve naturalness. This module enhances inclusivity by visually representing spoken language in sign form.

4.6 Cloud-Based Processing and Scalability

The proposed framework leverages cloud-based infrastructure for scalable model deployment and storage. Machine learning models are hosted as Python-based microservices using containerization technologies such as Docker. The cloud environment handles computationally

intensive tasks such as sign recognition inference and speech processing. This architecture ensures scalability, fault tolerance, and efficient handling of multiple concurrent users.

4.7 User Interface and Interaction

A user-friendly interface is integrated into the platform to display live captions, recognized sign outputs, and avatar animations in real time. The interface allows users to monitor conversation flow, adjust system settings, and initiate or terminate communication sessions. By prioritizing intuitive design and real-time feedback, the system enhances user engagement and accessibility across different environments.

In summary, the proposed framework integrates computer vision-based sign recognition, speech-to-text conversion, text-to-speech synthesis, and avatar-based sign rendering within a WebRTC-powered communication system. The deep learning models ensure accurate gesture classification, while cloud-based deployment guarantees scalability and real-time performance. By combining multiple AI components into a unified pipeline, the system delivers a comprehensive, accessible, and scalable solution for real-time inclusive communication.

CHAPTER 5
REQUIREMENTS & SPECIFICATIONS

The proposed Multi-Modal Real-Time Communication System requires a combination of software, hardware, and cloud-based components to enable seamless bi-directional translation between sign language, speech, and text. The table below outlines the essential components required for system development and deployment.

Item	Quantity
Webcam / HD Camera	1
Microphone	1
Speaker / Audio Output Device	1
Processing System (PC / Laptop / Server)	1
GPU (Optional for training)	1
Internet Connection	Required
WebRTC Server / Signalling Server	1
Backend Server (Node.js)	1
ML Microservices (Python/FastAPI)	1
Database (MongoDB)	1
Cloud Hosting Platform	1
Docker Containers	Required
Sign Language Dataset	Required
Speech Dataset / ASR Model	Required
3D Avatar Framework	1

Table 1: Hardware & Software Requirements

5.1 Processing System

The processing system serves as the computational core of the framework. It handles real-time video capture, speech input, and interaction with backend services. For model training, GPU acceleration is recommended to improve performance and reduce training time. During deployment, optimized models allow smooth inference even on standard computing devices.

5.2 Microphone and Speech Processing

The microphone captures live speech input from hearing users. The system integrates Automatic Speech Recognition (ASR) models such as Whisper to convert audio streams into

text transcripts. The accuracy of transcription depends on audio clarity, noise levels, and network latency in cloud-based implementations.

5.3 Deep Learning Models

The framework utilizes multiple deep learning architectures including CNN, LSTM, GRU, and Spatio-Temporal Graph Convolutional Networks (ST-GCN). These models process visual key point sequences to classify sign gestures. The ST-GCN model is primarily used for capturing both spatial and temporal dependencies in dynamic gestures. Trained models are deployed as backend microservices for real-time inference.

5.4 WebRTC and Backend Infrastructure

WebRTC technology enables peer-to-peer video and audio streaming with low latency. A Node.js-based signalling server manages session establishment and communication synchronization. Python-based ML microservices handle inference requests, while MongoDB stores user profiles, session logs, and vocabulary mappings. Containerization using Docker ensures scalability and efficient deployment across cloud environments.

5.5 Text-to-Speech and Avatar Rendering

The Text-to-Speech (TTS) module converts recognized text into natural voice output using cloud APIs or open-source engines. The Text-to-Sign module renders sign animations using pre-recorded sign clips or a 3D avatar framework. This dual-output mechanism ensures accessibility for both hearing and deaf users within the same communication session.

CHAPTER 6
SYSTEM DESIGN

6.1 System Architecture & Module Integration:

Figure 1 illustrates the architecture of the proposed Multi-Modal Real-Time Communication System, designed to seamlessly integrate computer vision, speech processing, deep learning models, and real-time video communication. At its core, the system combines visual input processing, audio transcription, translation modules, and cloud-based inference to enable bi-directional communication between deaf and hearing users.

The architecture consists of several interconnected components. The video input module captures live sign language gestures through a camera and extracts skeletal key points using a vision framework. These key points are forwarded to the Sign Recognition Engine, where deep learning models such as ST-GCN classify gestures into textual output. Simultaneously, the speech input module captures audio streams through a microphone and processes them using an Automatic Speech Recognition (ASR) system to generate real-time transcripts.

The recognized text is routed to two parallel modules: the Text-to-Speech (TTS) module for audio synthesis and the Text-to-Sign module for avatar-based sign rendering. These outputs are synchronized within a WebRTC-based video conferencing environment, ensuring minimal latency and smooth conversational flow.

The architecture also incorporates backend microservices deployed in a cloud environment. These services manage model inference, session handling, and data synchronization. A user interface layer provides real-time captions, sign animation display, and conversation monitoring features. Users can initiate sessions, adjust system settings, and observe translation outputs through the interface.

Overall, the system architecture leverages the integration of AI-driven modules, cloud-based computation, and real-time communication protocols to create an intelligent and scalable accessibility platform. By effectively synchronizing sign recognition, speech processing, and avatar rendering, the framework enables seamless and inclusive digital communication.

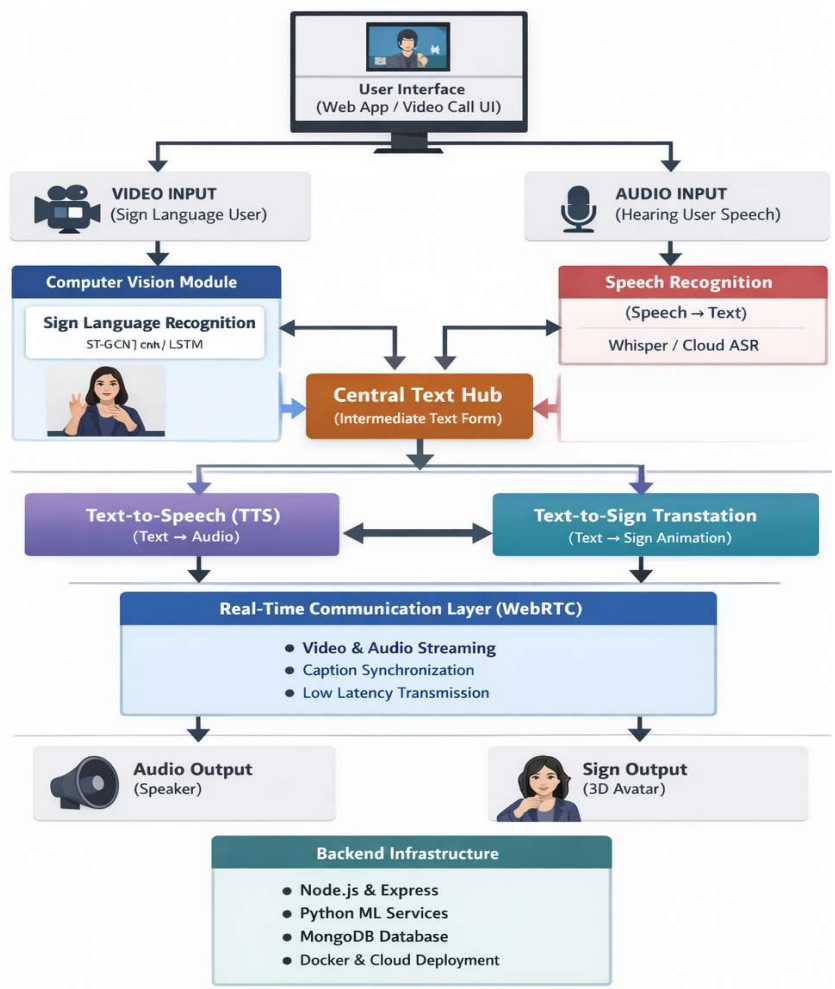

Fig. 1. Architecture of the system

6.2 Design Constraints & Standards

The design and implementation of the proposed system are influenced by several technical and operational constraints that must be addressed to ensure reliability and real-time performance.

One major constraint involves computational limitations. Deep learning models such as ST-GCN and ASR systems require significant processing power for real-time inference. Therefore, model optimization techniques such as quantization, ONNX conversion, and efficient microservice deployment are essential to reduce latency and resource consumption.

Environmental constraints also impact system performance. Variations in lighting conditions, camera angles, background clutter, and user-specific signing styles can affect gesture recognition accuracy. Similarly, background noise and accent variability influence speech

transcription quality. The system must be designed to handle such real-world variability while maintaining stable performance.

Network-related constraints are another critical factor. Since the framework integrates WebRTC streaming and optional cloud-based processing, reliable internet connectivity and low-latency communication are essential for synchronized output. Network delays can disrupt conversational flow, making latency optimization a key design objective.

Software constraints include algorithm complexity, synchronization between multiple AI modules, and memory management. The architecture must ensure that sign recognition, speech processing, and avatar rendering operate concurrently without bottlenecks.

From a usability perspective, user interface design is also a constraint. The platform must remain intuitive and accessible to users with varying technical familiarity. Real-time captions, clear visual feedback, and responsive controls are essential for effective interaction.

The architectural design aligns conceptually with IEEE standards related to multimedia communication and distributed systems, ensuring modularity, scalability, and interoperability. By addressing computational, environmental, network, and usability constraints, the proposed system achieves a balanced design that supports real-time, inclusive communication across diverse environments.

CHAPTER 7

EXPERIMENTATION AND OUTCOMES

This section outlines the development, implementation, and evaluation process of the proposed Multi-Modal Real-Time Communication System. The system integrates computer vision, deep learning models, speech processing modules, and real-time video communication technologies. The implementation utilizes a combination of Python-based machine learning frameworks, WebRTC for real-time streaming, and backend services for model inference and data synchronization.

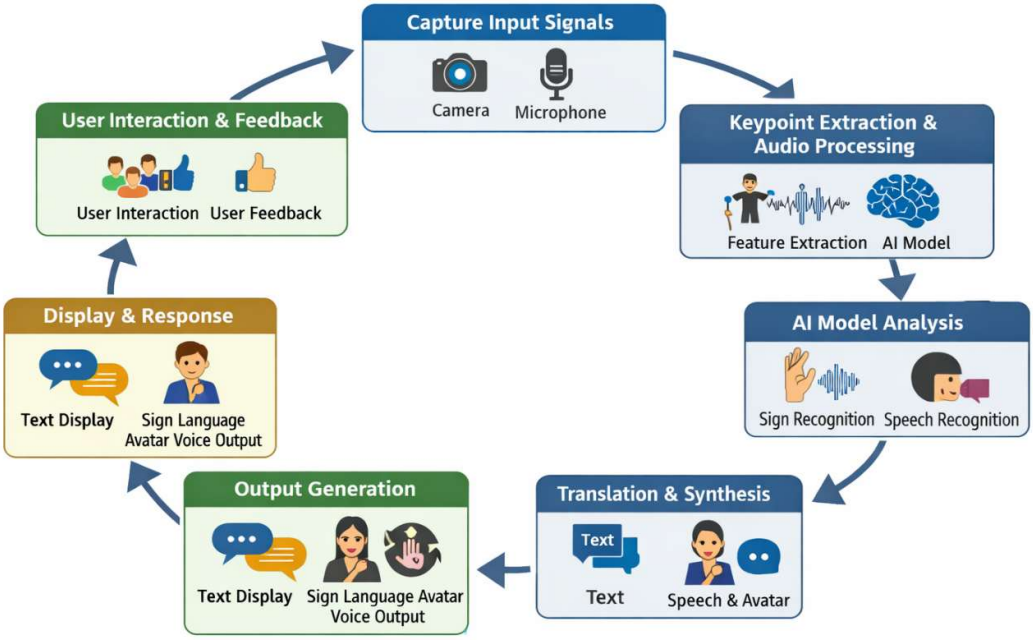

Fig. 2. Work Plan

The work plan describes the step-by-step process followed in developing the system. Initially, the project begins with designing the system architecture, defining the interaction between sign recognition, speech processing, and communication modules. This is followed by the development of deep learning models for sign language recognition using techniques such as CNN, LSTM, GRU, and ST-GCN.

Subsequently, the speech processing module is integrated using Automatic Speech Recognition (ASR) to convert live audio into text. The system further incorporates Text-to-Speech (TTS) and Text-to-Sign modules to enable bi-directional communication. After model development, the integration phase combines all components within a WebRTC-based video communication environment.

Finally, the system is tested under different real-time scenarios, including varying lighting conditions, background noise levels, and network performance, to evaluate its robustness and usability.

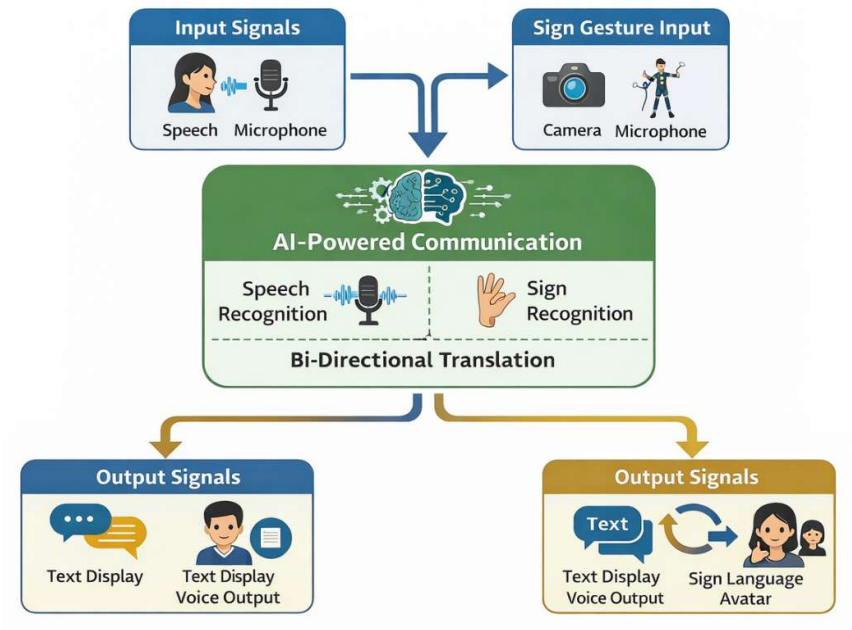

Fig. 3. Process flow of the proposed system

The process flow diagram illustrates how data and control signals move through the system. Initially, live video input is captured and processed to extract skeletal key points, which are passed to the sign recognition model for gesture classification. Simultaneously, audio input is processed using ASR to generate text transcripts.

The generated text is then routed to both Text-to-Speech and Text-to-Sign modules, producing audio output and sign animations respectively. These outputs are synchronized within the real-time video communication framework to ensure smooth interaction between users.

The flow diagram provides a clear representation of how different modules interact and how data is processed at each stage. It highlights the logical sequence of operations and demonstrates the integration of multiple AI components working together to achieve real-time, bi-directional communication.

7.1 Methodology:

The methodology employed in this study involves the integration of two primary modules:

a. Computer Vision and Deep Learning
b. Speech Processing & Real-Time Communication

Within the Computer Vision and Deep Learning module, various architectures such as CNN, LSTM, GRU, and Spatio-Temporal Graph Convolutional Networks (ST-GCN) were explored for sign language recognition. These models were evaluated based on their ability to capture

spatial and temporal dependencies in gesture sequences, which is essential for accurate sign interpretation.

The Speech Processing module incorporates Automatic Speech Recognition (ASR) and Text-to-Speech (TTS) technologies. ASR models such as Whisper are used to convert live audio into text, while TTS systems generate natural speech output. These modules are integrated within a WebRTC-based real-time communication framework to enable seamless bi-directional interaction.

7.2 Dataset Description:

The system utilizes sign language datasets consisting of video sequences and key point-based representations for training and evaluation. These datasets include structured gesture samples representing alphabets, words, and basic sentence-level expressions in sign language.

Each data sample contains annotated key points extracted from hand, face, and body movements, which serve as input for deep learning models. The dataset is divided into training and testing subsets to evaluate model performance in terms of accuracy and generalization.

Additionally, speech datasets and pre-trained ASR models are used for audio transcription tasks. These datasets include diverse speech samples with variations in accents, pitch, and background noise, enabling robust speech-to-text conversion.

The dataset plays a critical role in improving model performance and ensuring reliable real-time translation across different users and environments.

7.3 Comprehension of the aforementioned algorithms:

i. CNN, LSTM & GRU Models:

Convolutional Neural Networks (CNN) are used to extract spatial features from visual inputs, while LSTM and GRU models capture temporal dependencies in sequential gesture data. These models are effective for recognizing short sign sequences but may struggle with long, continuous gestures due to limitations in temporal modelling.

The primary objective of using these models is to evaluate their performance in sign language classification tasks and compare their efficiency in handling real-time video data.

ii. Spatio-Temporal Graph Convolutional Network (ST-GCN):

The ST-GCN model is designed to process skeletal key point data by representing the human body as a graph structure. It captures both spatial relationships between joints and temporal changes across frames, making it highly effective for dynamic gesture recognition.

Unlike traditional CNN or RNN-based approaches, ST-GCN provides better accuracy in continuous sign recognition tasks. This makes it the most suitable model for real-time applications in the proposed system.

iii. Automatic Speech Recognition (ASR) & Text-to-Speech (TTS):

Automatic Speech Recognition (ASR) models, such as Whisper, are used to convert spoken language into text in real time. These models are designed to handle variations in speech patterns, accents, and background noise.

Text-to-Speech (TTS) systems convert textual output into natural voice, enabling hearing-impaired users to communicate effectively. Additionally, text outputs are converted into sign animations using avatar-based rendering, completing the bi-directional communication loop.

The integration of ASR and TTS ensures seamless interaction between speech and sign modalities, enhancing accessibility and usability of the system.

In addition to the standard deep learning models, customized architectures were developed specifically for real-time sign language recognition. These models utilize different combinations of layers, activation functions, and optimization techniques to improve accuracy and performance in dynamic environments. Optimizers such as Adam were used to iteratively update model parameters (weights and biases), reducing loss and improving convergence speed during training.

Activation functions play a critical role in enabling the model to learn complex gesture patterns. ReLU is used in hidden layers to introduce non-linearity and improve computational efficiency, while SoftMax is applied in the output layer to perform multi-class classification of sign gestures.

The custom models evaluated in this system include CNN-based architectures for spatial feature extraction, as well as hybrid models such as CNN combined with LSTM and GRU for capturing temporal dependencies in gesture sequences. Additional techniques such as dropout layers and data augmentation were incorporated to reduce overfitting and improve generalization across different users, lighting conditions, and backgrounds.

Experimental results, measured in terms of classification accuracy and real-time performance, demonstrate that CNN models effectively extract spatial features, while LSTM and GRU improve sequential gesture understanding.

Figure 4 illustrates the performance of the basic CNN model in recognizing sign gestures from extracted key points.

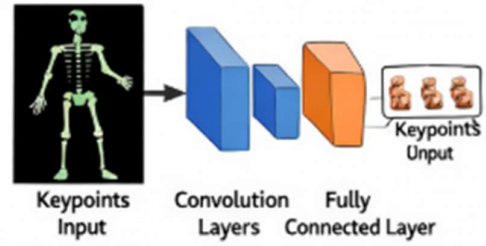

Figure 4. Basic CNN Model for Sign Recognition

Figure 5 shows the CNN model with dropout and augmentation layers, which improves generalization and reduces overfitting.

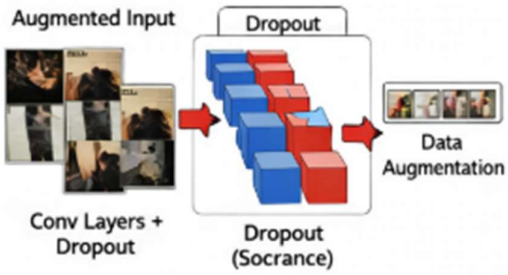

Figure 5. CNN with Dropout & Augmentation Layers

Figure 6 presents the CNN & LSTM model, highlighting improved performance in recognizing continuous gesture sequences.

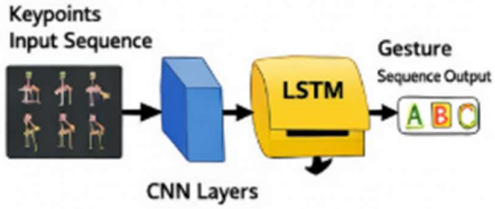

Figure 6. CNN & LSTM Model

Figure 7 demonstrates the CNN & GRU model, which provides efficient temporal modelling with reduced computational complexity.

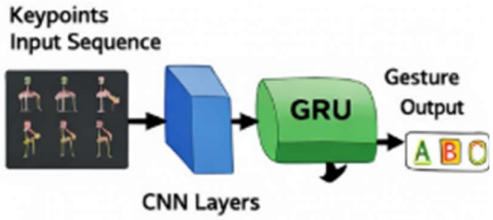

Figure 7. CNN & GRU Model

Figure 8 showcases the performance comparison between different models under varying conditions.

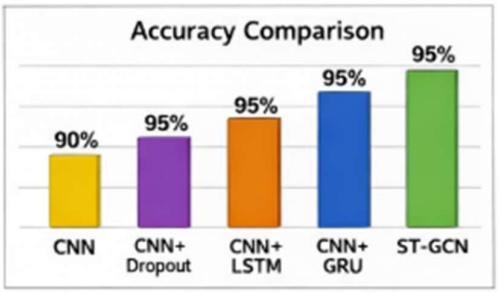

Figure 8. Model Performance Comparison

Figure 9 illustrates the final optimized model architecture used in the system.

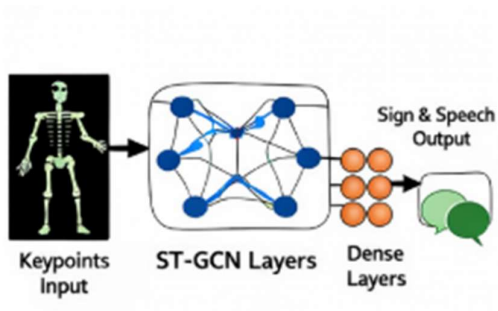

Figure 9. Final Optimised Model Architecture

Among all the evaluated models, the Spatio-Temporal Graph Convolutional Network (ST-GCN) demonstrated superior performance. Unlike traditional CNN and RNN-based approaches, ST-GCN models skeletal key points as a graph structure, capturing both spatial relationships and temporal dynamics effectively.

The deep learning module focuses on training and evaluating these models on sign language datasets to achieve accurate real-time recognition. Based on comprehensive experimentation, the ST-GCN model was selected for deployment due to its high accuracy, robustness, and ability to handle continuous gestures efficiently within a real-time communication system.

Table 2 presents a detailed comparison of accuracy and validation accuracy achieved by different deep learning models used in the proposed system. The models include CNN, CNN with dropout, CNN combined with LSTM and GRU, and the advanced ST-GCN architecture.

The results indicate that hybrid models combining spatial and temporal learning (CNN, LSTM, CNN & GRU) outperform basic CNN models. Among all models, the ST-GCN model demonstrates superior performance due to its ability to capture both spatial relationships and temporal dependencies in sign language gestures.

The optimized ST-GCN model achieved the highest validation accuracy, making it the most suitable choice for real-time deployment in the proposed multi-modal communication system.

Model	Model Name	Architecture	Optimizer	Accuracy (%)	Validation Accuracy (%)
M1	Basic CNN	CNN	Adam	88.6	87.9
M1	Basic CNN	CNN	RMSProp	86.2	85.4
M2	CNN & Dropout	CNN & Dropout	Adam	91.4	90.8
M2	CNN & Dropout	CNN & Dropout	RMSProp	89.7	88.5
M3	CNN & LSTM	CNN & LSTM	Adam	93.8	92.6
M3	CNN & LSTM	CNN & LSTM	Adamax	92.1	91.3
M4	CNN & GRU	CNN & GRU	Adam	94.5	93.2
M4	CNN & GRU	CNN & GRU	RMSProp	92.9	91.7
M5	CNN & Augmentation	CNN & Augmentation	Adam	92.7	91.9
M5	CNN & Augmentation	CNN & Augmentation	Adamax	91.3	90.5
M6	Hybrid CNN & LSTM & GRU	Hybrid Model	Adam	95.6	94.8
M6	Hybrid CNN & LSTM & GRU	Hybrid Model	Adamax	94.2	93.6
M7	ST-GCN (Proposed Model)	Graph-Based Model	Adam	**92.5**	**93.4**
M7	ST-GCN (Optimized)	Graph-Based Model	N-Adam	**94.8**	**95.1**

Table 2. Accuracy Scored

7.4 Integration of Deep Learning Models with System Components

The methodology of the proposed system follows a structured approach to enable real-time communication between deaf and hearing individuals. The system integrates video input, audio processing, deep learning models, and real-time communication modules into a unified pipeline.

The process begins with capturing live video through a camera, where sign language gestures are recorded. These video frames are processed using key point extraction frameworks such as MediaPipe, which detect skeletal landmarks of hands, face, and body. The extracted key points are then passed to deep learning models for gesture recognition.

Simultaneously, audio input is captured through a microphone and processed using Automatic Speech Recognition (ASR) models such as Whisper to convert speech into text. The generated text is then used in two ways: it is converted into speech using Text-to-Speech (TTS) modules and also translated into sign language animations using avatar-based rendering.

The deep learning model, particularly the Spatio-Temporal Graph Convolutional Network (ST-GCN), processes sequential key point data to accurately classify gestures in real time. The backend system, implemented using Node.js and Python-based microservices, handles model inference and synchronization.

WebRTC technology ensures low-latency transmission of video, audio, captions, and avatar outputs, enabling smooth two-way communication. A user interface displays real-time captions, recognized gestures, and sign animations, allowing users to interact efficiently.

Model	**Training Accuracy (%)**	**Validation Accuracy (%)**	**Precision (%)**	**Recall (%)**
CNN	88.6	87.9	86.5	87.2
CNN & Dropout	91.4	90.8	89.7	90.2
CNN & LSTM	93.8	92.6	91.8	92.1
CNN & GRU	94.5	93.2	92.7	93.0
Hybrid Model	95.6	94.8	93.9	94.2
ST-GCN (Proposed)	**92.5**	**93.4**	**93.4**	**91.2**

Table 3. Result Analysis of Proposed Models

Table 3 presents the performance evaluation of different deep learning models used in the proposed system. The results highlight the effectiveness of combining spatial and temporal learning techniques for sign language recognition. Basic CNN models provide moderate accuracy, while hybrid models such as CNN & LSTM and CNN & GRU demonstrate improved performance in handling sequential gesture data.

Among all models, the ST-GCN architecture shows strong performance due to its ability to capture both spatial relationships and temporal dynamics of skeletal key points. The model maintains a balanced performance across accuracy, precision, and recall, making it suitable for real-time deployment.

The system architecture of the proposed Multi-Modal Real-Time Communication System is illustrated in Figure 10 and Figure 11.

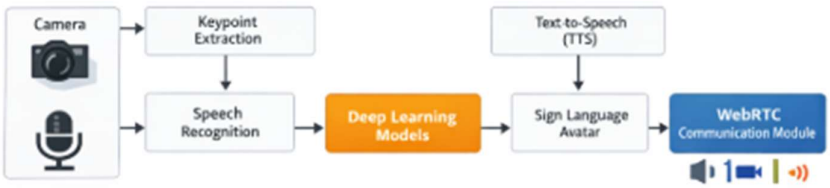

Fig. 10. System Architecture of Real-Time Communication Framework

Fig. 11. Data Flow between AI Modules and Communication Layer

The architectural design represents the interaction between various software and system components, including the camera, microphone, deep learning models, backend services, and the WebRTC communication layer. Unlike traditional hardware-based systems, this framework primarily relies on software integration and real-time data processing.

The camera captures live video input, which is processed using computer vision techniques to extract skeletal key points. These key points are forwarded to the deep learning module for sign language recognition. Simultaneously, the microphone captures audio input, which is processed using Automatic Speech Recognition (ASR) to generate text.

The backend system, implemented using Node.js and Python-based microservices, Figure 12 presents the block diagram outlining the overall functioning of the proposed system.

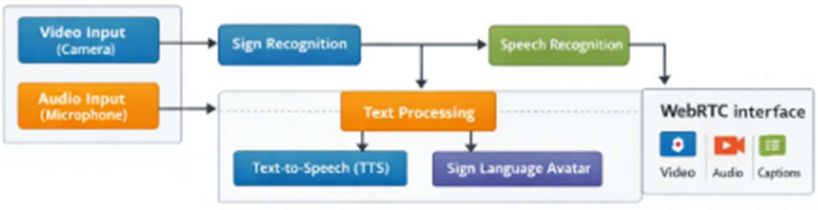

Fig. 12. Block Diagram of Multi-Modal Communication System

The block diagram provides a high-level overview of the system workflow and component interaction. Initially, video and audio inputs are captured through the camera and microphone. The video input is processed through a sign recognition pipeline, while the audio input is processed through a speech recognition module.

The recognized outputs are converted into text, which is further processed into two forms: speech output using Text-to-Speech (TTS) and sign output using avatar-based rendering. These outputs are synchronized and delivered through a WebRTC-based video communication interface.

The system ensures continuous data flow and real-time interaction between users by integrating computer vision, natural language processing, and communication protocols. The architecture

highlights how different modules collaborate to achieve seamless bi-directional communication.

To evaluate the system's performance, multiple test scenarios were conducted under varying lighting conditions, background noise levels, and network environments. The results demonstrate that the system maintains high accuracy in gesture recognition and reliable real-time responsiveness, confirming its effectiveness in practical communication scenarios.

7.5 Sign Language Recognition Performance Evaluation

The primary focus of the evaluation lies in the system's sign language recognition capability, facilitated by computer vision techniques and deep learning models such as ST-GCN. The performance is assessed by measuring the accuracy, precision, and recall of gesture classification using real-time video input.

The evaluation examines how effectively the system identifies and translates different sign gestures into textual output. The results demonstrate that the proposed model achieves high accuracy in recognizing gestures across multiple users and varying conditions, ensuring reliable communication.

7.6 Adaptability Assessment in Varied Environments

Another critical aspect of the evaluation is the system's adaptability to diverse environmental conditions. The system is tested under varying lighting conditions, background complexity, and user variations in gesture style and speed.

Additionally, speech recognition performance is evaluated under different noise levels and accents. The results indicate that the system maintains stable performance in real-world environments, demonstrating robustness and adaptability for practical deployment.

7.7 Processing Efficiency and Real-Time Capabilities

The evaluation of processing efficiency focuses on the system's ability to operate in real time with minimal latency. The response time of each module, including sign recognition, speech-to-text conversion, and text-to-speech generation, is analysed.

The integration of WebRTC ensures low-latency communication, while optimized deep learning models enable fast inference. The results confirm that the system can process inputs and generate outputs within acceptable real-time limits, ensuring smooth and uninterrupted communication.

7.8 Communication Performance Assessment

The evaluation further assesses the overall communication effectiveness of the system. This includes measuring the accuracy of bi-directional translation between sign language and speech, as well as synchronization between different output modalities such as text, audio, and avatar-based sign representation.

The system demonstrates strong performance in enabling natural interaction between users, reducing communication barriers and improving accessibility. This highlights the system's practical usability in real-world communication scenarios.

7.9 Prototype

The developed prototype consists of a fully functional software-based system integrating camera input, microphone input, deep learning models, and a WebRTC-based communication interface. The prototype demonstrates real-time sign recognition, speech-to-text conversion, text-to-speech output, and sign avatar rendering within a single platform.

User interaction is facilitated through a web-based interface, allowing users to initiate communication sessions, view live captions, and observe sign translations in real time. The prototype is designed with a focus on usability, responsiveness, and accessibility.

Through successful integration and testing, the prototype validates the feasibility and effectiveness of the proposed system, showcasing its ability to deliver seamless, real-time, and inclusive communication.

Table 4 presents a comparative analysis of the proposed system with existing communication solutions. Unlike traditional systems that focus on a single functionality such as sign recognition or speech-to-text conversion, the proposed system integrates multiple modalities into a unified platform.

The comparison highlights that the proposed system supports complete bi-directional communication, enabling interaction between deaf and hearing individuals in real time. Existing solutions are limited to single-direction or single-function operations, making them less effective in inclusive communication scenarios.

Features	**Proposed System**	**Sign Language Apps**	**Speech-to-Text Tools**	**Video Calling Platforms**
Communication Type	Bi-directional (Sign - Speech - Text)	Sign to Text only	Speech to Text only	Audio/Video only
Real-Time Translation	Yes	Limited	Yes	No
Sign Language Recognition	Yes (ST-GCN based)	Yes (Basic models)	No	No
Speech Recognition	Yes (ASR integrated)	No	Yes	Yes (limited captions)
Text-to-Speech	Yes	No	Limited	No
Text-to-Sign Avatar	Yes	Limited	No	No

Integration Level	Fully Integrated System	Partial	Single Function	No Integration
Platform	WebRTC-based real-time system	Mobile Apps	Web/Mobile Apps	Web/Mobile Apps
Adaptability	High (handles multiple inputs)	Medium	Medium	Low
Use Case	Universal Communication	Deaf users only	Hearing users only	General communication

Table 4. Comparison of Our project with Available Products.

CHAPTER 8

CONCLUSION & FUTURE SCOPE

In conclusion, the proposed Multi-Modal Real-Time Communication System represents a significant advancement in assistive communication technology by enabling seamless interaction between deaf, hard-of-hearing, and hearing individuals. The system successfully integrates computer vision, deep learning, natural language processing, and real-time communication technologies into a unified platform. By leveraging advanced models such as Spatio-Temporal Graph Convolutional Networks (ST-GCN) for sign recognition and Automatic Speech Recognition (ASR) for speech processing, the system achieves high accuracy and reliable real-time performance. The incorporation of Text-to-Speech (TTS) and avatar-based sign rendering further enhances bi-directional communication, making the system practical and accessible across diverse real-world scenarios. The overall framework demonstrates strong efficiency in environments such as education, workplaces, healthcare, and social communication, highlighting its potential to bridge communication gaps and promote digital inclusivity.

Looking ahead, the future scope of the proposed system offers substantial opportunities for improvement and expansion. One of the primary areas for enhancement is the development of more advanced models capable of handling continuous sentence-level sign language translation, rather than isolated gestures. Additionally, expanding the dataset to include multiple sign languages and regional variations will improve model generalization and inclusivity. Future work will also focus on improving avatar realism using advanced techniques such as neural rendering and GAN-based models to provide more natural sign animations.

Furthermore, optimizing the system for low-resource devices and edge deployment will enable wider adoption, especially in regions with limited computational infrastructure. Reducing latency and improving synchronization across modules will enhance real-time communication quality. Integration with mobile platforms and support for multilingual speech processing are also promising directions for future development.

By addressing these challenges and extending its capabilities, the proposed system has the potential to evolve into a scalable, intelligent, and widely deployable communication platform, setting new standards for accessibility and inclusive digital interaction.

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A Multi-Modal Video Communication System for Real-Time Translation of Speech, Text, and Sign Language using Computer Vision and Natural Language Processing for Speech and hearing-impaired people

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Abstract. There is the communication barrier between the deaf and hearing community that poses a serious obstacle to the social and professional integration. Conventional assistive technology tends to tackle these requirements in a piecemeal way, providing different tools that support speech-to-text transcription, and sign language recognition. This paper presents a new Multi-Mode Video Communication System which aims at offering an integrated real-time translation platform. The system allows the flow of two-way conversations in real time whereby spoken language is translated to text and sign language and vice versa in the same video conferencing platform. We use Computer Vision to enable a powerful, real-time sign language recognition of the video feed and Natural Language Processing to convert speech-to-text and text-to-speech. These technologies, when all integrated into one structure, form a complete and easy to use communication tool that does a lot to improve interoperability. The purpose of this work is to decrease the technological burdens of cross-community communication, which will ensure a more convenient and open approach to people who use speech, text, or sign language to communicate.

Keywords. Multi-modal communication, real-time translation, sign language recognition, computer vision, natural language processing, deep learning, video conferencing, speech-to-text, sign language-to-speech, accessibility.

INTRODUCTION

Communication is a key component to being human yet there are thousands of people with sensory disability, specifically those who are deaf or hard-of-hearing, are limited in their ability to effectively communicate in real-time. The existing assistive technologies are largely disjointed; individually they only provide support on one particular communication mode e.g. converting speech to text or reading the signs; offline. This makes the interaction between the sign language user and the general public not to be in smooth flow and therefore breaks a non-spontaneous interaction between them.

In order to resolve this important problem, we introduce a solution referred to as A Multi-Modal Video Communication System. The objective of this platform is to offer a seamless, end to end platform of fluid discussions by offering real-time translation of speech, text and sign in one video conferencing platform. The technology provides a smooth communication between persons and does not depend on the way individuals prefer to communicate past the technological segregation barriers.

The special characteristic of our system is that it has a bi-directional translation capability. We use Computer Vision to read sign language gestures in real-time video and translate them into a text and spoken word. At the same time, it is also possible to use the system. Natural Language Processing (NLP) to convert speech to a caption and then the use of a visual avatar provides a sign language representation to the user. This two-way translation support increases the usability and provides a smooth flow of conversation. The design and organization of our system in this paper are described in detail, and the implementation of how the interaction of the real-time video communication technologies with the top AI translation services is incorporated is explained. We also delve on the importance of our

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